

# VivaGuard - Backend Architecture & Endpoint Documentation

## 1. Executive Summary & Overview

VivaGuard backend is a FastAPI Python web application providing real-time AI technical interview simulations, thesis defense grading, voice agent text-to-audio synthesis, and WebSocket streaming capabilities.

## 2. Backend File Structure & Component Roles

|                                        |                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| app/main.py                            | FastAPI app instance creation, CORS middleware setup, and api_router inclusion.                                    |
| app/core/config.py                     | Loads environment variables (.env) such as API keys, host, port, and rate limits via Pydantic.                     |
| app/api/deps.py                        | Dependency injection container providing SessionBroker, RateLimiter, and AssemblyAIClient singletons.              |
| app/api/v1/api.py                      | Main API router combining all endpoint routers into a unified v1 routing tree.                                     |
| app/api/v1/endpoints/health.py         | Endpoint GET /health to verify backend service uptime.                                                             |
| app/api/v1/endpoints/test_interview.py | Endpoints POST /test-interview/generate, /evaluate-question, /evaluate-audio, and /cumulative-report.              |
| app/api/v1/endpoints/tts.py            | Endpoints POST /api/v1/tts and /voice-agent/text-to-audio for text-to-speech audio synthesis.                      |
| app/api/v1/endpoints/debrief.py        | Endpoint POST /api/v1/debrief generating post-defense STAR framework reports.                                      |
| app/api/v1/endpoints/websockets.py     | WebSocket handler /ws/copilot & /ws/live for real-time audio streaming & signaling.                                |
| app/services/assemblyai_stt.py         | AssemblyAI Speech-to-Text transcriber for both audio blobs and streaming mic audio.                                |
| app/services/assemblyai_tts.py         | AssemblyAI Voice Agent Text-to-Audio generator converting text into PCM/WAV speech audio.                          |
| app/services/test_simulator.py         | LLM-driven technical question generator, response scoring engine (0-100), and adaptive difficulty tier calculator. |
| app/services/evaluator.py              | STAR framework defense session evaluator analyzing filler words, clarity, depth, and examiner sentiment.           |
| app/services/broker.py                 | WebSocket session broker managing subscriber connections and broadcasting live evaluation events.                  |
| app/services/rate_limiter.py           | Token bucket rate limiter enforcing request rate limits across endpoints.                                          |

## 3. API Endpoints Declaration vs Frontend Usage

| Method | Endpoint Path                            | Declared In (Backend)       | Used In (Frontend)     | Purpose                        |
|--------|------------------------------------------|-----------------------------|------------------------|--------------------------------|
| GET    | /health                                  | endpoints/health.py         | Status Checkers        | Health & uptime check          |
| POST   | /api/v1/test-interview/generate          | endpoints/test_interview.py | TestSimulatorPhase.tsx | Generates questions            |
| POST   | /api/v1/test-interview/evaluate-question | endpoints/test_interview.py | TestSimulatorPhase.tsx | Grades text response           |
| POST   | /api/v1/test-interview/evaluate-audio    | endpoints/test_interview.py | TestSimulatorPhase.tsx | Transcribes & grades audio     |
| POST   | /api/v1/test-interview/cumulative-report | endpoints/test_interview.py | TestSimulatorPhase.tsx | Cumulative readiness scorecard |
| POST   | /api/v1/tts                              | endpoints/tts.py            | TTS Components         | Voice Agent Text-to-Audio      |
| POST   | /api/v1/debrief                          | endpoints/debrief.py        | DebriefPhase.tsx       | STAR defense report            |
| WS     | /ws/copilot                              | endpoints/websockets.py     | useAudioStreamer.ts    | Realtime WebSocket stream      |